

3 The ventilation rate of animals will depend on many factors.

(a) State **two** lung conditions that affect the ventilation rate of a human.

(2)

.....

.....

(b) The table shows the mean resting ventilation rate of five species of animal.

Animal	Body mass / kg	Resting ventilation rate / breaths per second
Talapoin monkey	1.3	0.54 ± 0.09
Ring-tailed lemur	2.9	0.53 ± 0.04
Crested porcupine	20.0	0.30 ± 0.03
Siberian tiger	134.0	0.27 ± 0.06
Hippopotamus	2210.0	0.12 ± 0.03

(i) Comment on the relationship between the mean resting ventilation rate and the body mass of animals.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

(ii) The tidal volume for a resting talapoin monkey was found to be 40 cm^3 .

The minute ventilation is the volume of gas inhaled or exhaled from the lungs in one minute.

Calculate the respiratory minute ventilation for a resting talapoin monkey.

Give your answer in dm^3 to **two** significant figures.

(2)

..... dm^3

(c) An increase in activity level results in a change in the ventilation rate.

Describe why this change occurs and how it is controlled.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 3 = 11 marks)